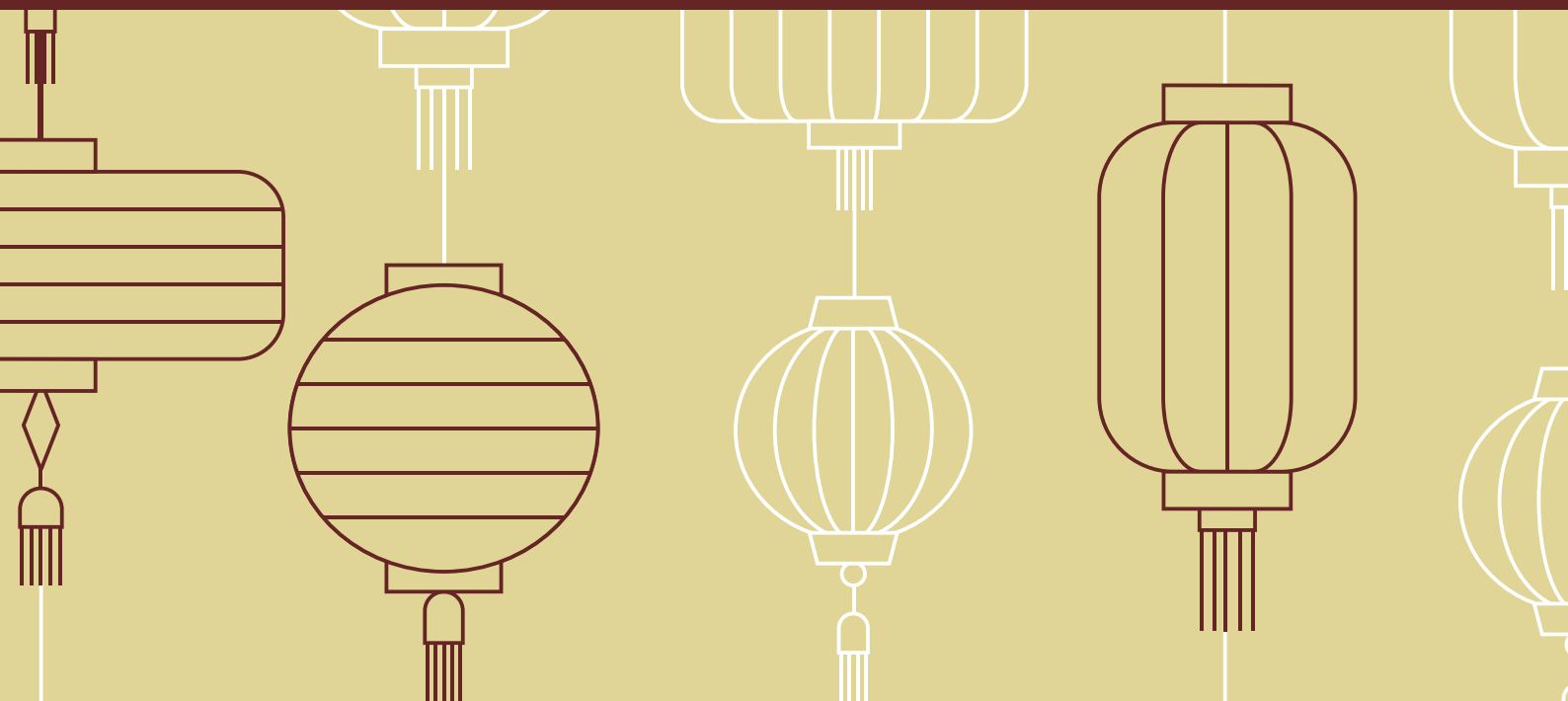

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1) Wireless Com

2) A Satellite Communication Basically Consist of two Parts
   1) Satellite space
   2) Ground station

A Satellite receivers receives the signal transmitted from the ground station in the form of uplink frequency.
It is proceed by the Satellite to enhance the strength of signal & the transmitt back to the station in the form of downlink
Processing of signal is done by Radio repeaters known as transponders

The first concept of Satellite is given by English man Arthur C. Clark in 1945

The first Satellite launched by USSR. which is known as

* This achievement was followed by, in 1958 by American Artificial Satellite which is used to Broadcast the message by the President on the Christmas eve

Two Satellite were deployed first a reflector Satellite called echo. & second is transmitter Satellite called

In 1962 active Communication Satellite called Telstark & Relay were launched

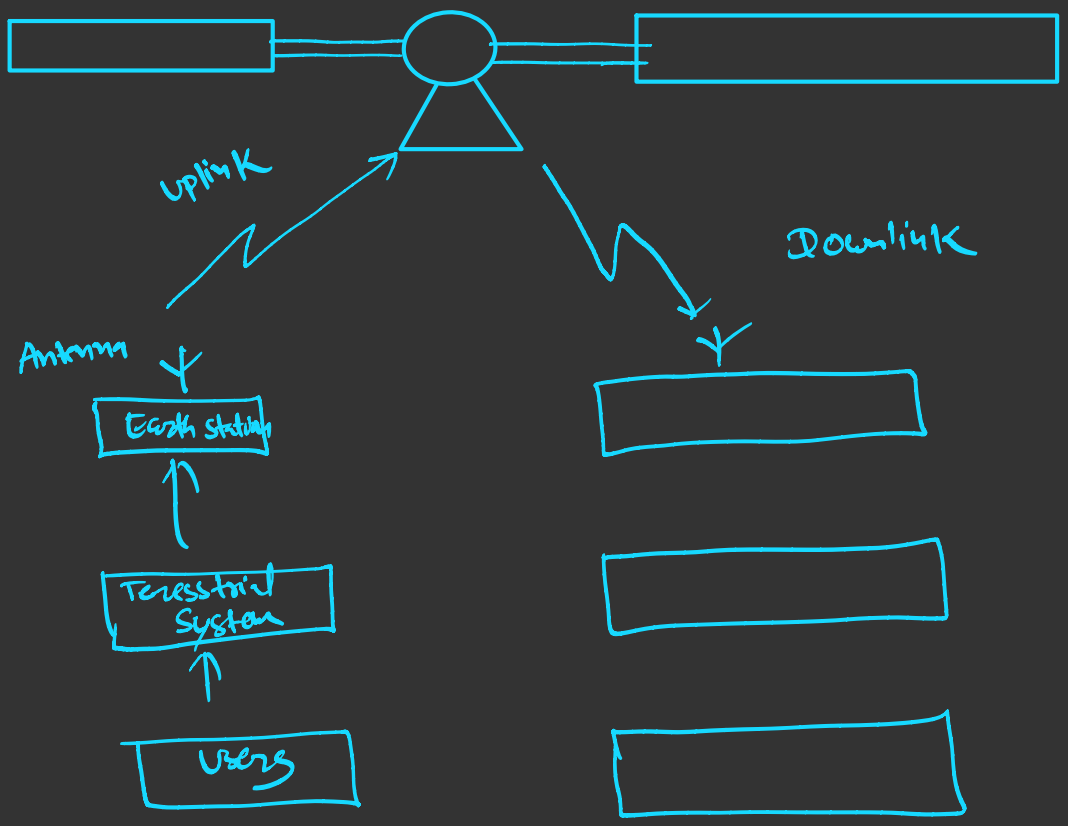
- The 1st Geostationary Satellite was launched in 1963
- What is Geo stationary Satellite → Imp for exam
  - ↳ where is it used
  - ↳ Advantages / disadvantages
- In 1965 1st commercial Geostationary Satellite intelsat 1 was inaugrated after that series of intersat Satellite were launched
- The 1st Communication Satellite was launched by Soviet Union. This Satellite provide video & voice Communication Simultaneously

Basic Concept of Satellite Communication

- A Communication Satellite is a orbiting Artificial Earth Satellite that receives a Communication signal from a transmitting ground station amplifies & possibly process it then transmit it back to the Earth for reception by 1 or more receiving ground station

A Satellite Communication Utilize EM wave to carry the information from the ground to Space & back to the Earth

- An EM wave consist of electrical & Magnetic field that are ⊥ to each other & to the direction of Propagation
- Element of Satellite Communication



The basic element of communication satellite services are divided b/w 1) Space segment
2) Ground segment

1) Space
Consist of spacecraft, launch mechanism

2) Ground
Comprises the Earth stations & Network control center of entire satellite system

Space segment

The Satellite itself is also known as space segment & is compose of 3 seperate Units namely fuel system, the satellite, telemetry Controls & the transponders

- The Transponders includes the receiving antenna to pickup signals from the ground station, a broad band receiver, A i/p multiplexer & a frequency converter which is used to reroute the receive signal through a high power amplifier for download

- The primary role of a satellite is to reflect electronic signal. In the case of telecon satellite the primary task is to receive signal from a ground station & send them back to another ground station located at a considerable distance away from the 1st.
  This relay action can be two way as in the case of long distance Phone Call, Another use of satellite is when for the television Broadcast

- The use of satellite is observed where the satellite is equipped with Camera & Various sensors

Ground

- This is the Earths segment, uplink, downlink, terrestrial signal is passed through a Base band Processor, an up Converter, highly Power amplifier through a Parabolic antenna.

2) In case of downlink or receiving station works in the reverse process as the uplink section proceeds.
   The received signal is received by the Parabolic antenna & is Convert into a Baseband signal.

Assignment - 1

i) Give the brief introduction about satellite communication along with the background historical overview

ii) write the basic Concept of satellite communication, along with the elements used in satellite comm

- Draw the proper diagram of satellite communication explain every blocks in detail

iii) Define freqency band used for satellite communication

iv) define polarization in terms of satellite communication

v) List different types of satelite along with the services they are used

→ In word file

Types of Satellite

1) **fixed satellite** services

It is used for point to point communication
ex- links for existing telephone networks

2) Broadcast telephone service

ex- Satellite telephone & Radio

3) Mobile service satellite

Used for satellite phone, Aeronautical & Marine time mobile

4) Global Positioning technique (GPS)

used for metrological satellite services. (weather monitoring

5) Deep space Satellites.

* Satellite Orbits

The satellite can be rotated around the earth through various paths these paths are called orbits of the satellite.
These orbits are used to cover the specific range & application area

Geosynchronous

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* Geostationary satellite

* Geothermal satellite

* Geosynchronous Orbit

When the inclination of the orbit is not 0° . It is called Geosynchronous orbit. The period of Geosynchronous orbit is equal to period of revolution of earth with itself. This means the Geostationary

* Diff b/w Geosynchronous orbit

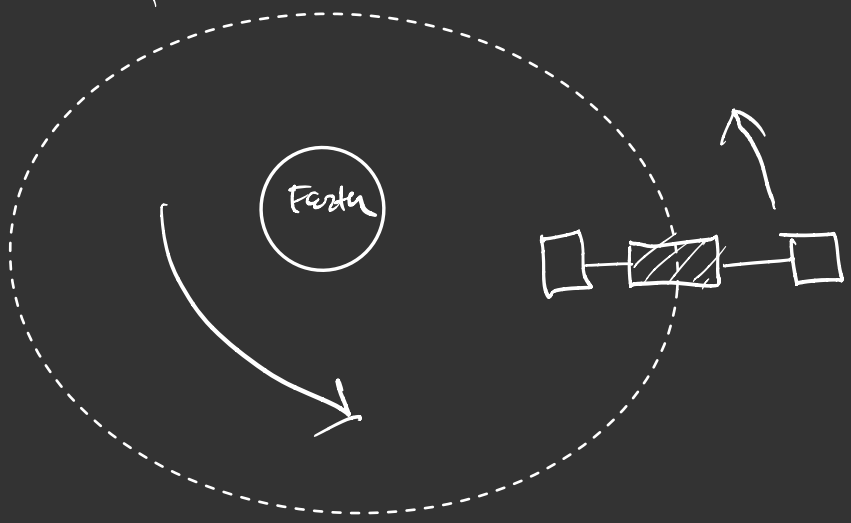
The orbit is circular	i) This orbit is not circular
ii) This orbit lie in the equatorial plane if inclination is 0°	ii) This orbit is inclined w.r.t to equatorial plane
iii) A satellite in this orbit appears to be stationary w.r.t Earth	ii) A satellite in this orbit appears to be oscillating w.r.t point on earth
iv) The time period of satellite in this orbit is 23hr 56 min & 4.1 sec	iv) Same time period
v) There is only one Geostationary orbit	v) There can be many Geosynchronous orbit

Orbit Shapes

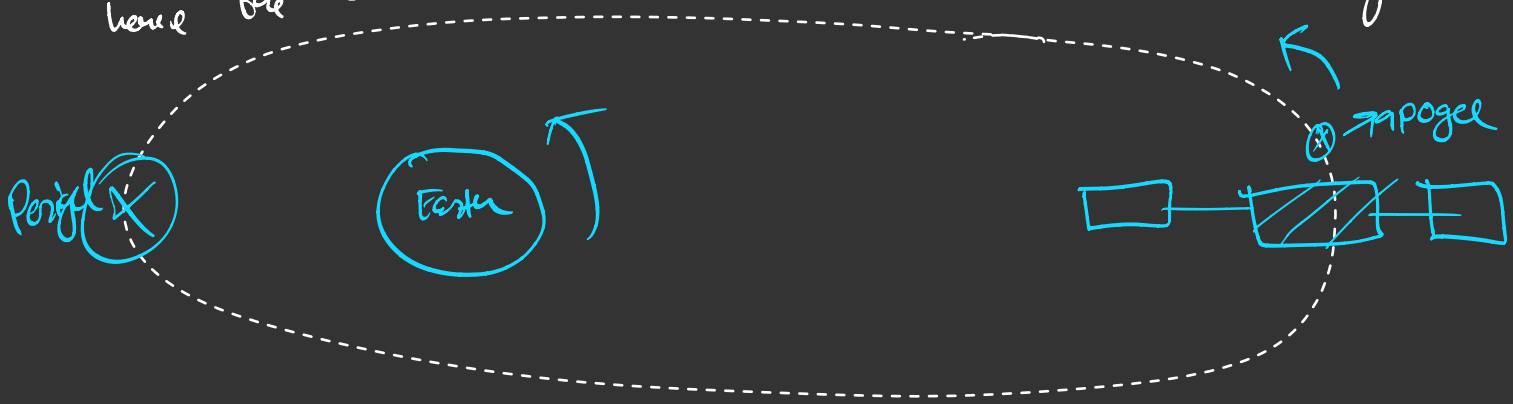
The orbit shape itself can be

ii) For circular path the height of the satellite is the distance of satellite from the Earth. But for calculation purpose the height is distance b/w center of Earth & satellite

Circular satellite



Elliptical orbits are inclined at an angle of 64° w.r.t to the equatorial plane for elliptical path center of earth is one of the focal points of ellipse hence the distance b/w earth & satellite varies during the revolution



The point in the orbit where the satellite is closest to the earth is perigee.

& the point where the satellite is farthest from the earth is known as apogee

The distance of the perigee & apogee are measured from

Therefore apogee & perogee includes the Radius of the Earth.

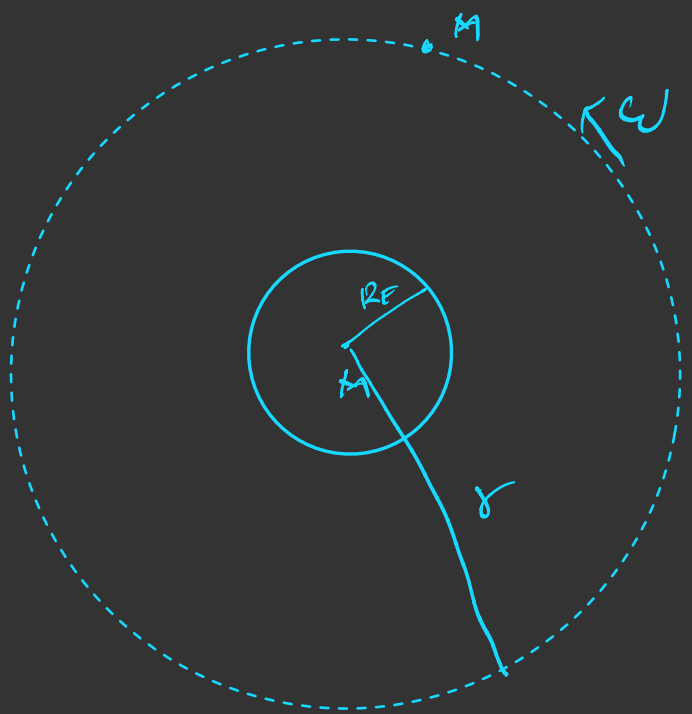
Height of a Geostationary orbit

$$\frac{GMm}{r^2} = mr\omega^2$$

At the surface of the Earth, gravitation force is mg
& At the surface of the earth is $\frac{GMm}{R_E^2} = mg$

$$\Rightarrow R_E = \sqrt{\frac{GM}{g}}$$

$$\frac{GMm}{r^2} = mr\omega^2$$



$$r = \left(\frac{gR_E^2}{\omega^2}\right)^{1/3}$$

$$= \left(\frac{gR_E^2}{4\pi^2 R^2}\right)^{\frac{1}{3}}$$

if f = 1 Rotation / day, R_E = 6370 km

$g = 9.9\ m/s^2$

$r = 43000...$

Therefore height, h of a geostationary satellite over the Earth surface is $h = r - R_E$

$= 35038$ km

* Geostationary Satellite

Geostationary or Geosynchronous satellite orbits the earth around the Equator in a circular way with a angular velocity equal to that of earth.

Geostationary satellite travels in the same direction as the Earth rotation. Since the satellite revolves exact synchronous with the earth rotation.

Therefore it appears fixed or stationary to a observer on earth.

Therfore such satellite are called as Geostationary or Geosynchronous satellite

- Geostationary satellite remains apparently fixed that is satellite is continously visible from point on

Therfore no special tracking antenna associated are required

The fixed antenna pointing towards satellite can be used in keeping in track for a satellite is relatively easy

- The effect of doppler shift in frequency results due to change in relative movement b/w source & Receiver.
- For Geostationary satellite the angular velocity V, is given by $V = r\sqrt{\frac{g}{r+h}}$; ∵ R is the average Radius of earth.
  g is the gravitational acceleration.
  h -> height of satellite above ground.

* Advantage of Geostationary

Keeping track of Geostationary is relatively easy as the satellite remains almost stationary in respect to given a station.

ii) The relative posn of sat ... & earth station are fixed hence continuous communication is possible by 1 satellite only. That is no need to switch from one satellite to another

iii) There is no break in communication as only 1 satellite is to track

iv) The coverage area on the earth is very large because the height of satellite is more

v) Very small energy storage is required As at very high altitude of satellite it is coming under high intensity sol

vi) The effect of doppler shift freq is almost zero.

The advantage of Geostationary Satellite

i) because of higher altitude of Satellite

ii) Propagation time of signal is

iii) Signal has to travel longer distance

iv) Attenuation takes place highly sensitive receivers are required.

iii) There is practically inability to service Geostationary

5) highly precise spe

Application of Geostationary Satellite

Most Common Satellite are Geo stationary Satellite because for TV Broadcasting, Regional, National & Global Common

iii) T

iv) Mobile T

v) private networks for corporation &

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* Station keeping

A Geostationary Satellite must lie in equatorial plane be circular & have correct Altitude

Once a satellite is placed in

central force tends to deviate it from required orbit. Thus drift in position is undesirable —

For reliable Contacts

to be esta

ii) Actually the satellite is subjected to force other than earth gravitational force such as gravitational force of sun & moon, also the gravitational force of earth is not uniform at all points on the earth because the earth is not a perfect sphere instead it is flattened at both poles.
It is necessary to adjust the orbit of satellite periodically

iii) Accurate prediction of satellite posn one or 2 week ahead required skilled specimen ship. An orbit Control
The task of correcting deviation is referred to as Station Keeping.

The principal task of Station keeping is to sense Satellite posn parameter & provide corrective

various sensing devices such as radiation sensory device & infrared sensors installed on Satellite board

iv) Non Geo stationary orbit: Satellite can be categorised depending on their altitude above the earth surface.
They are categorised as - Geo stationary earth orbit
ii) Medium earth orbit - MEO
iii) low earth orbit - LEO

The geostationary satellite altitude is about 36000 km
MEO " " " " 10000 - 15000 km
LEO " " " " confined around 500 - 15000 km

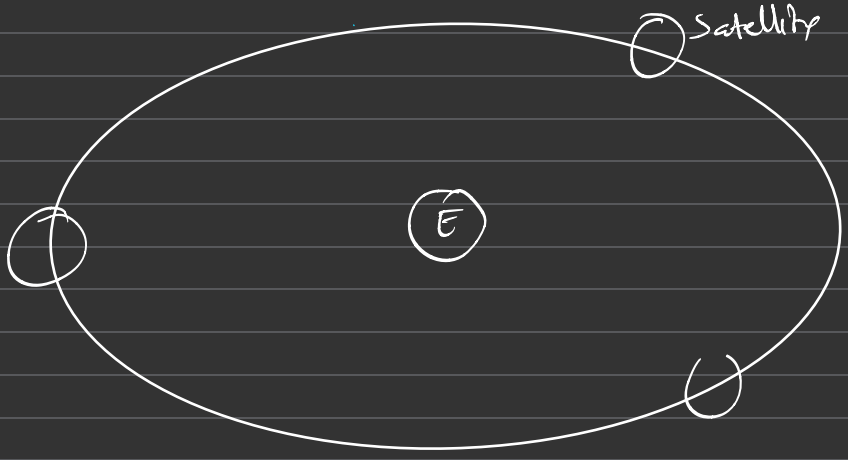
MEO & LEO satellite are referred to as non-geo stationary satellite

i) Geo stationary Earth orbit

Geo is also called Geo synchronous or synchronous orbit which has 23 hr 56 min 4.09 sec or 24 hr revolution time. but are inclined w.r.t Equator. orbits that are below mean altitude

ii) orbit that below mean altitude of about 35784 km have prior revolution shorter than 24 hrs & hence are termed as non-geo.

iii) Geo has the ability to provide coverage of entire hemisphere at one time
Satellite are designed to only to last 15 years on orbit because of particles in ability to serve a satellite in geo & replenish consumable
A system of 3 satellites in geo each separated by 120° of longitude can receive & send of/ over all parts of the world



iii) The effect of doppler shift

doppler shift in frequency results due to change in relative movement b/w source & Receiver for a geostationary

Satellite the orbital velocity V is given by

$$V = R\sqrt{\frac{g}{R+H}}$$

* Medium Earth Orbit

The altitude of MEO is around 1500-36000 km
- Satellite orbit is about 6 hrs which is much longer than LEO Satellite tends to hack over
- Transmission are greater than LEO but still significantly less than GEO.

- Advantage of MEO
MEO Satellite can provide truly global
- MEO sat offers lower path loss than geo.
- Propagation delay is moderate. ie. 30 - 100 ms depending on the height
- The satellite architecture is distributed therefore it gives robustness
(v) lower path loss make it possible to use hand held terminals

Disadvantages

1) To cover entire globe large no. of satellites are required typically 10-12
2) Received signal strength is variable because of Range
3) doplar effect can be observed sign
4) Techno is yet to matured
5) It increases orbital due to large no. of Satellite per system
6) Initial devel of such syste requires large time several months to year

Applica

i) Surveillance of earth surface

ii) GPS

iii) Data gathering for military

LEO

The iridium mobile satellite systems in which satellite are at altit of approx 1000 km & each passes a given user in only a few seconds.

ii) The adv of using non-geo satellite is smaller & hence less radiated power is required & the prop delay is reduced as well.

iii) in a seamless way Sat orbit period is b/w 1.6 to 1.8hr. the higher period of orbit resulting in slightly longer revolution

iv) Non geo Sat of alt of about 1500 km are subjected to atmospheric drag & are likely to required replacement after 5 to 8 years

Adv

Can provide true global coverage

ii) offers lowest path loss compared to LEO & GEO

iii) Propagation delay is lowest & is comparable

iv) Leo sat used freq spectrum effectively.

Disadvantages

1) To cover entire globe large no. of satellite are required
2) dopler effect are its highest
3) archit of space segments is most complex
4) Rate of depositing orbital is highest
5) since the satellite replacement rate is more than geo & leo maintain

App

- Used for mobile satellite comm
- Used in earth surface surveillance
  " " " resource application
- " " geological survey

Comparison of LEO, MEO & GEO

Parameter	LEO	MEO	GEO
Delay (in ms)	low (15	Medium (70)	High (280
link diversity	possible	possible	limited
satellite power	low	medium	High
Satellite Mask	low	medium	High
Doppler frequency	High	medium	very low
No. of satellite for Global coverage	40	10–80	3
Hand off rate	High	Medium	nony
Ground segment complexity	High	High	Low
System cost	High	med high	med
Advantages	i) Smaller handset ii) less Req power iii) low delay time iv) Freq Reuse suitable for positioning	i) less handoff than leo ii) less propagation delay than geo	i) Can cover almost entire world using 3 satellite ii) Can continously monitor one point on earth surface iii) Good for broadcasting
Disadvantage	i) Require large no. of of Satellite ii) iii) multiple satellite iv) atmospheric drag	i) More Satellite Req than Geo. ii) Greater delay & Propgation losses than leo	i) Can not cover high altitude or low elevation ii)

Orbital mechanism

1) The orbital loc of the spacecraft in a comm satellite, systems plays a major role in determining the coverage & operational characteristics of the service provider by that system.

2) Newton's law of motion governs artificial earth satellite that orbit the earth. Satellite orbit determination based on laws of motion deployed by Kepler & later refined by Newton

3) Competing force acts on the satellite gravity tends to pull the satellite in towards the earth while its orbital velocity tends to pull the satellite away from the earth

The gravitational force, $F_{in} = m(\mu/r^2)$

angular velocity force $F_{out} = m\left(\frac{V^2}{r}\right)$

$$V = \sqrt{\frac{\mu}{r}}$$

This gives the velocity required to maintain a satellite at the orbit

μ = kepler constant $= 3.986 \times 10^5$ km^3/s^2

r = radius from earth

m = satellite mass

Kepler's law

i) Satellite is a artificial object located by rocket in space orbit following the laws in it motion. As the planet rotating around the sun

ii) Kepler's law of planetary motion applied to any two bodies in space that interface through gravitation. The law of motion are described through 3 fundamental principles derived by Keplers as follows:

1) Kepler 1st law VVI

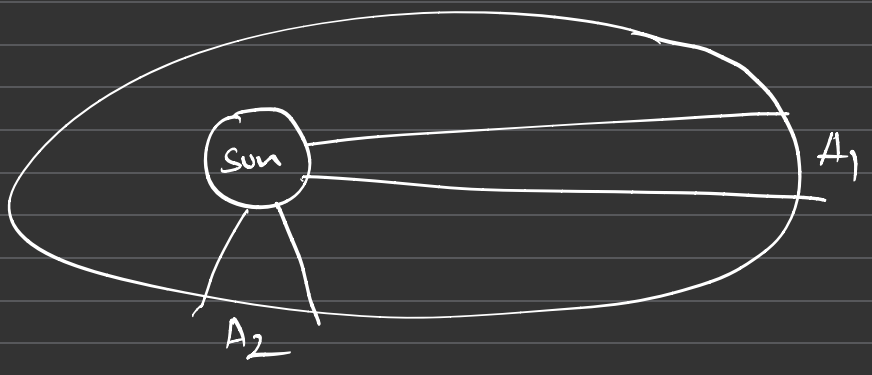
The orbit of each planet follows an elliptical path in space with the sun as 1 focus motion lies in the plane around the sun

explanation:- The path followed by the Satele around the primary is ellipse

The ellipse has two focal points. The centre of mass is always centred at one of the foci.

Kepler's 2nd law:-

The line from the sun to the planet or radius vector out equal area in equal interval of time. This is known as the law of area.



$A_2 = A_1$

Kepler's 3rd law:

The square of the Planet orbital period around by sun is proportional to the cube of semi-major axis of the ellipse. For a planet in a solar system

$$T^2 = \left[\frac{4\pi^2}{\mu}\right] a^3$$

The ... proportional to mean ... of distance b/w two bodies

$\therefore T^2 \propto a^3$

T is orbital period in 2nd.

$a \rightarrow$ distance b/w 2 bodies

$\mu \rightarrow$ Kepler's Constant

if orbit is circular

$a = r$ [$\because r$ is radius]

$$T^2 = \left[\frac{4\pi^2}{\mu}\right] r^3$$

$$r = \left[\frac{\mu}{4\pi^2}\right]^{T^{2/3}}$$

orbital radius is constant multiplied by orbit period multiplied $T^{2/3}$

Under this condn a speci ... only by proper selection of the orbit radius this allows the satellite designer to select orbit periods that meets particular application requirement by locating the satellite at proper orbit altitude

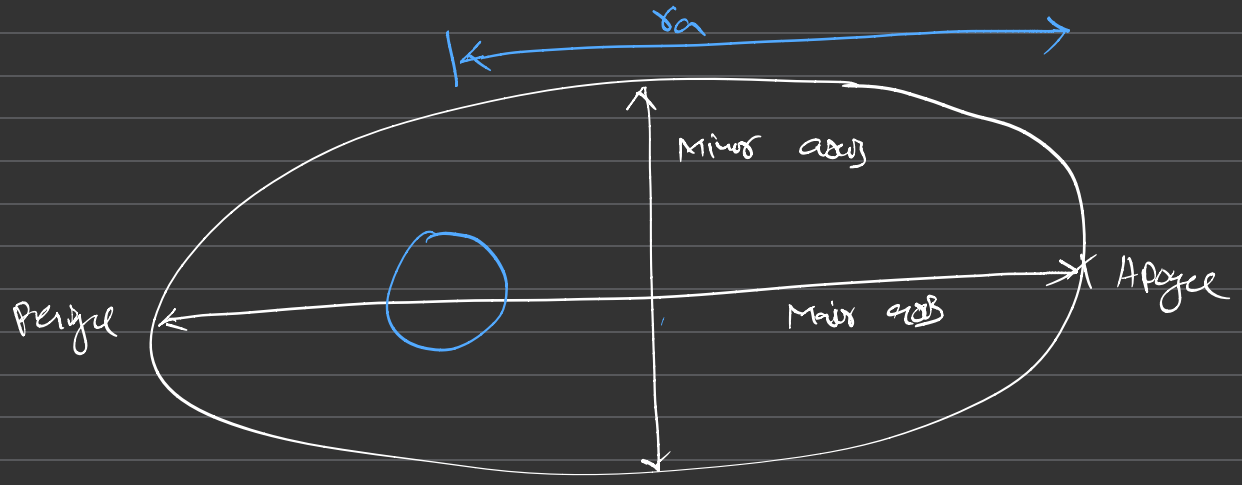
- from kepler's 3rd law the tangential orbital velocity [km/s] at radius r is calculated for a circular orbit from $V = \left(\frac{\mu}{r}\right)^{1/2}$
- A day is the time b/w consecutive crossing of any particular longitude on the earth with reference to inertial space that is in practice with reference to any star other than sun. this corresponds to a 360° rotation. A solar day is the time b/w consecutive crossing of any particular longitude by the sun / earth / as this corresponds to more than 360°

Orbital Parameters

- Apogee : The pt farthest from earth is known as Apogee.

Perigee: The point closest to the earth

line of : The line joining the Apogee & perigee is called as line of



The pt where the orbit crosses the equatorial plane going from south to North is called ascending node:

descending node:-

- line of nodes:- The line joining the ascending & descending nodes through the center of the earth.

- Argument of Perigee:- The angle from ascending node to Perigee. measured in the orbital plane.

write ex of the ascending nodes.

The angle measured eastward in the equatorial plane from the 1st to the ascending node. pt of Aries

The eccentricity is measure of the circularity of orbit. It is determined from

$$e = \frac{r_a - r_p}{r_a + r_p}$$

r_a → dist from center of earth to the apogee pt.

r_p → dist from center of earth to the perigee pt.

↳ eccentricity of the orbit

for elliptical objects, $0 < e < 1$

for circular $e = 0$

An almost endless no. of combⁿ of orbital parameter are available for satellite orbits
5 independent parameters can required to define an orbit size say an orientation in space & the fourth parameters define

Orbital elements

The set of parameters needed to uniquely specify the form of the orbital satellite the minm no. of parameter req is 6

These 6 parameter are orbital elements

1) Eccentricity of orbit (e)
2) semi-major axis of orbital ellipse (a)
3) Time of perigee defined by tp
4) Right ascension of ascending nodes
5) Inclination of orbits to equatorial plane (i)
6) Argument of perigee (ω)

Q. Calc the speed of a satellite in geo-synchronous equatorial circular orbit around the earth clockwise

with R = 4261

Q. A scientific satellite is moving on an elliptical orbit with perigee altitude of 400km & apogee altitude 15000 km. a) Calculate 'e' b) period of the this orbit

3. A satellite is in circular orbit. 322 km are in mts. i) orbital velocity ii) orbital period in minute iii) orbital angular velocity in rad/sec

4. Calc rad & height of Geo-orbit Given that sad day is 23 hrs 56 min & 4.09 sec

5. Considering the space shuttle orbit having height is equal 250 km. $r = 6378$ km obtain 1) orbital period 2) avg linear velocity.

6. Elliptical orbit perige = 1000 km. Apogee = 4000 km obtain a) period b) eccentricity

7. From a circular orbit calc the height of Geostationary satellite radius of earth = 6400 km

8. An earth obs satellite B in an inclined rectangular orbit and following orbital parameter semi-major axis 7830 km. $e = 0.01$ / Argument of perige = 180

Time of ascending node = 10hrs 30mm

Determine.
1) Radius of Apogee, r_a
2) r_p
3) orbital period
4) mean angular velocity
5) velocity at apogee